SQL FUNDAMENTALS:

> SELECT
> FILTERING

SECTION 1 SELECTING AND SORTING.

Data Source Table:

Table employees

Id	first-name	last_name	department	salary	hire_date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018	Durban
4	David	Nokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

Q1. Retrieve all columns from the employees table.

ANSWER.
```
SELECT *
FROM
    employees;
```

Table employees

id	First-Name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Nokoena	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2. Find all unique departments

Answer:
```
SELECT DISTINCT department
FROM
    employees;
```

Table employees

id	department
1	IT
2	HR
3	Finance
4	Marketing

Q3. Retrieve first and last names ordered by salary descending

Answer:
```
SELECT first_name, last_name
FROM Employees
ORDER BY salary Desc;
```

Table employees

id	first_name	last_name
1	Eva	Naidoo
2	Alice	Green
3	Cathy	Zulu
4	David	Mokoena
5	Brian	Lee

Q4. Retrieve the top 3 highest-paid employees

Answer:
```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

Table employee

id	first_name	last_name	~~Salary~~ salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

SECTION 2: FILTER WITH WHERE

Qs. Find employees in the IT department

ANSWer:
```
SELECT  first_name, last_name, department
From    employees
WHERE   department = 'IT';
```

Table: Employee

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6. Find employees in Finance with Salary > 60000

ANSWer:
```
SELECT  first_name, last_name, department
FROM    employees
WHERE   department = 'Finance'
AND     $ salary > 60000;
```

Table: employees

id	first_name	last_name	department
3	Cathy	Zulu	Finance

Q7. Find employees in HR or Marketing

ANSWer:
```
SELECT  first_name, last_name, department
From    employees
WHERE   department = 'Marketing'
OR      department = 'HR';
```

id	first_name	last_name	department
3	~~Cathy~~	~~Zulu~~	
42	Brian	Lee	HR
4	David	Mokoena	Marketing

Vindesh Gopaul

Q8. Find employees not in IT

Answer:
```
SELECT  first_name, last_name, department
From    Employees
WHERE   department  <> 'IT';
```

Table: employees

Id	first-name	last_name	department
2.	Brian	Lee	HR
3.	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Q9. Find employees in HR, IT, or Finance using IN.

ANSWER:
```
SELECT  first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Table: employees

Id	first_name	last-name	department
1.	Alice	Green	IT
5	Eva	Naidoo	IT
2.	Brian	Lee	HR
3	Cathy	Zulu	Finance

Q10. Find employees in IT with salary > 65000 and city Johannesburg

Answer:
```
SELECT  first_name, last_name, department, Salary city
FROM employees
WHERE   Salary > 65000
 AND    City = 'Johannesburg';
```

< Result table on next page>

Q 10. Continued.

Id	first_name	last_name	Department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naido	IT	72 000	Johannesburg